

* The arm. flux stationary wot poles.

$$\frac{62}{12}$$

$$\therefore \frac{I_a}{1} = \frac{50}{6} \times \frac{720}{2 \times 6}$$

$$AT/p$$

$$\frac{I_a}{4} \quad T = \frac{720}{2}$$

$$p = 6$$

$$AT/p = \frac{50 \times 720}{6 \times 2 \times 6} = 500 \text{ AT}$$

* Factors affecting terminal voltage of dc generator →

* When the gen^r is on NL the NL induced emf is E_0 . when it is loaded there is voltage drop due to arm. reaction demagnetisation.

$$E_0 - E_g = \text{Arm. reaction drop.}$$

* Due to arm. resistance there will be arm. resistance drop $I_a R_a$.

* These drops are proportional to load.

* In a separately excited generator there are only above two drops affecting the terminal voltage.

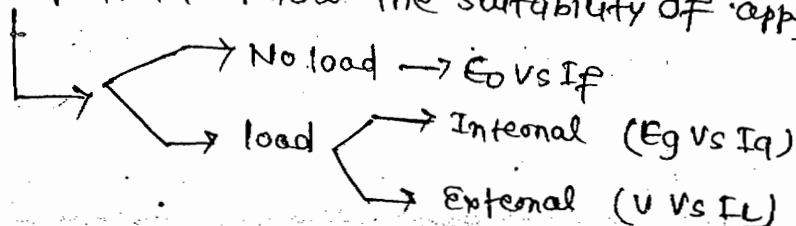
* In a shunt gen^r its excitation is the terminal vol.

Therefore reduction in V in turn reduces the terminal vol. itself.

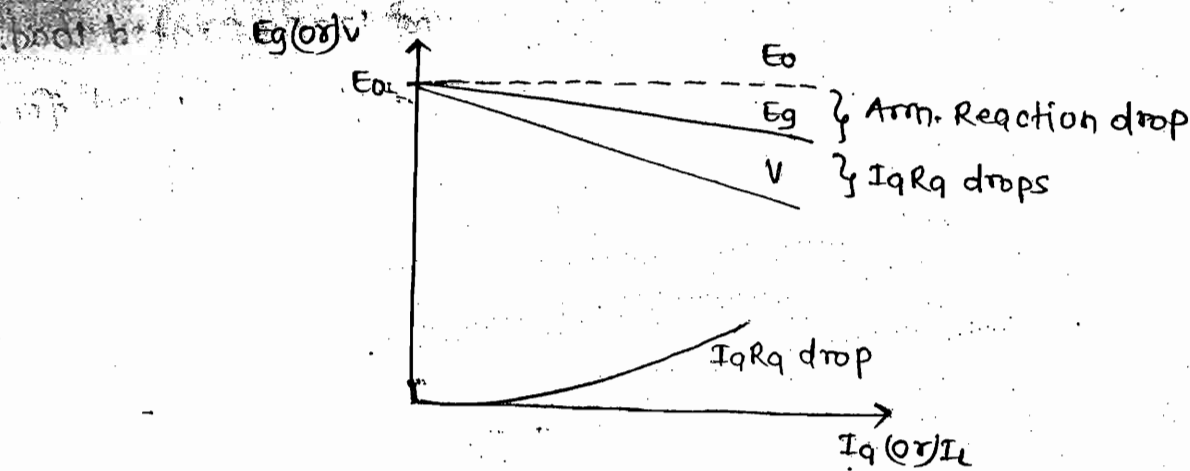
* This will be a cumulative effect when the gen^r is loaded beyond its rated value known as break down point.

* Characteristics →

* These are the graphical representation of the key parameters which are plotted to know the suitability of appln.

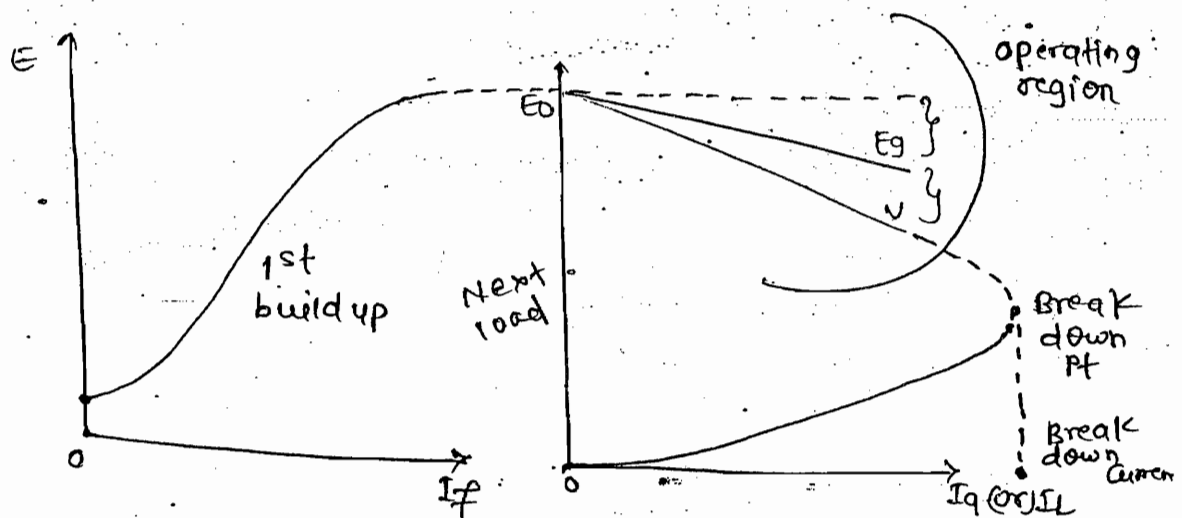


* Separately excited gen^r →



- * This are not commercially used for normal power supply because it requires additional dc. Vol. source.
- * It was used in excitation sys. of power plant gen^r.
- * Dc supplies in air crafts / ships.
- * Used in a speed control method known as Ward-Leonard method.

* Shunt gen^r →



- * In its operating region its flux remains approx. same, but beyond break-down value a cumulative reduction of vol. happen reduce the terminal vol. $\therefore$ drastically.

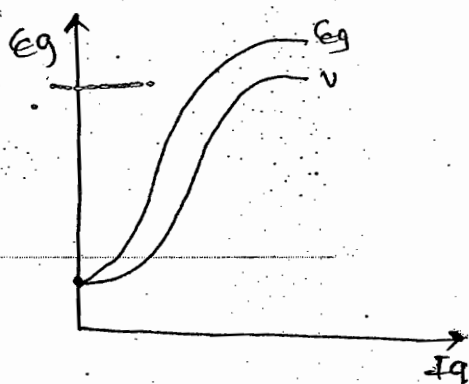
* Generally the operating region will be around 125% of rated load.

* It is used as small dc power supply & exclusively used for battery charging.

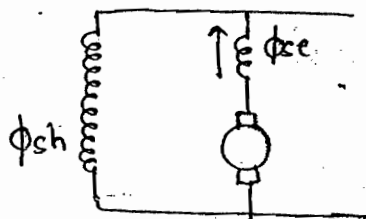
* It was also used in excitation sys. of power plant gen^r along with separately excited gen^r.

* Series Gen^r →

* As the field wdg is in series with arm. & load as the load increases flux increases to in turn increase the voltage. Therefore in its operating region it has rising vol. c/s also called as variable vol. gen^r. not suitable for ordinary power supplies but used as boosters in long feeders, particularly in dc distribution sys.



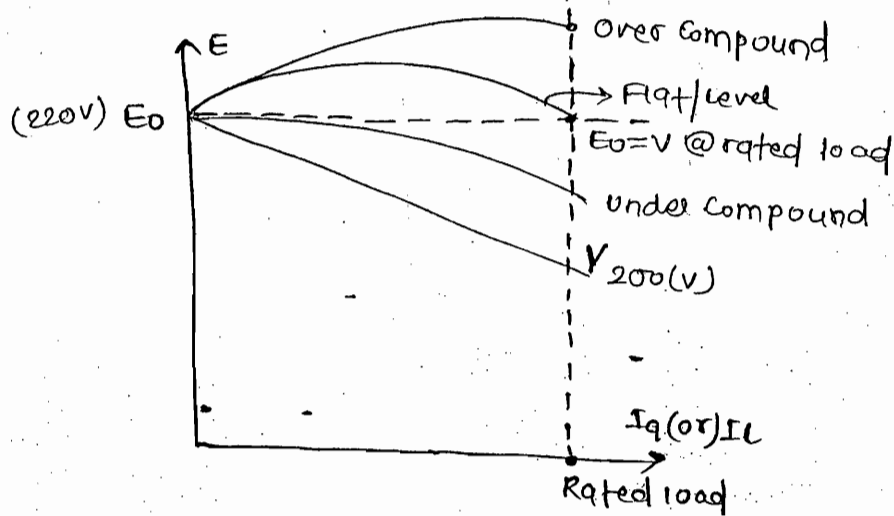
* Cumulative →



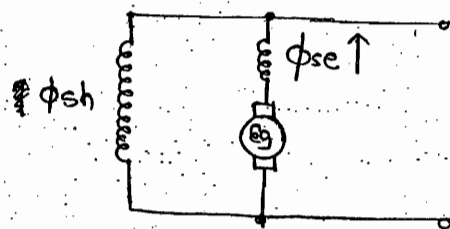
* As the series flux add its shunt flux the net flux increases with load. Consequently it has better vol. c/s. than separately & shunt gen^r.

It can be compounded to adjust its terminal voltage.

Therefore it is widely manufactured dc gen^r due to its flexible c/s which can be used in large rating dc power supplies.

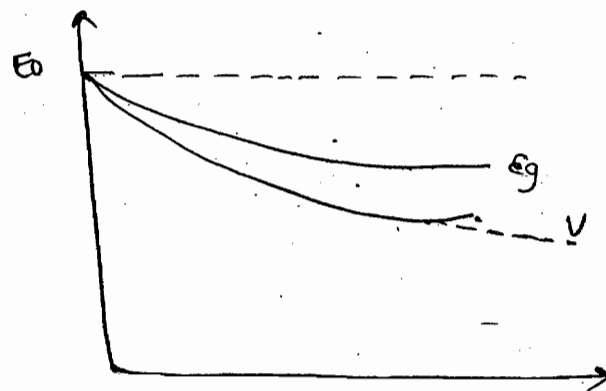


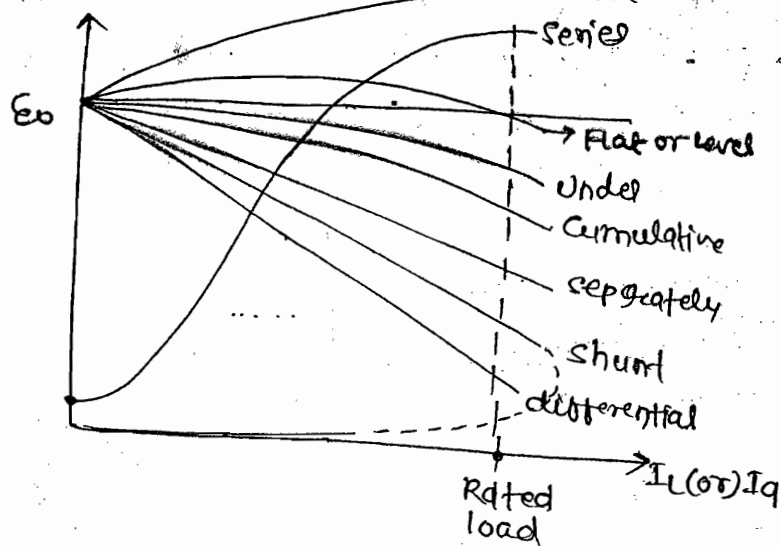
Differentially $\rightarrow$



* Arc weldings --

* The net flux decreases with load to decrease the terminal vol. This is not use for ordinary power supply but specifically used in welding purposes to limit the welding currents.





Voltage Regulation $\rightarrow$ * It is the change in terminal vol. when the full or rated load across the terminal is disconnected. keeping the flux & speed constant.

$$\% VR = \frac{E - V}{V}$$

$E \rightarrow$ NL induced vol.

$V \rightarrow$ Rated terminal vol. @ rated load

* VR is % drop in the m/c.

$$VR = \frac{V_{drop}}{V}$$

$V_{drop} \downarrow$ as possible

* It should be as min^m as possible, best/ideal value is 0. which happens for only flat/level compound gen.

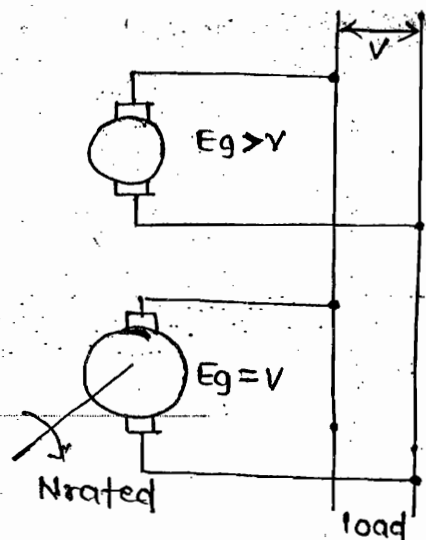
* For series & over compound VR becomes -ve. not suitable for ordinary load purpose.

* PARALLEL OPERATION →

Operating gen^r in parallel across the common terminals known as bus bars provide the following ad.:-

- (1) High ele inertia across the busbars (Constant voltage sys.)
- (2) High reliability.
- (3) Efficiency. (✓)
- (4) Future expansion.
- (5) Continuity of supply during maintenance & repairs.

Due to this reasons universally the gen^r in the power plants & all the power plants are operating in parallel to form a grid st.

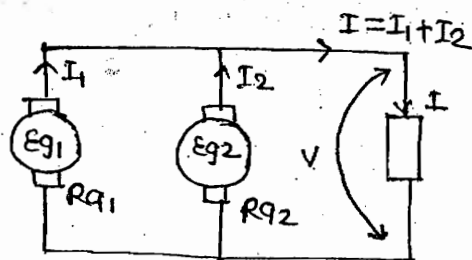


$E_g > V$	Generating mode
$E_g = V$	Floating mode
$E_g < V$	Motoring mode

* In order to connect 2 dc gen^r in parallel it requires 2 essential condⁿ:-

- (1) Terminal vol. should be same.
- (2) Polarity should be matched.

* Consider 2 dc gen^r operating in parallel at a common terminal vol. V & sharing a common load with induced emfs E_{g1} , E_{g2} respectively.



$$E_{g1} - I_1 R_{a1} = V$$

$$E_{g2} - I_2 R_{a2} = V$$

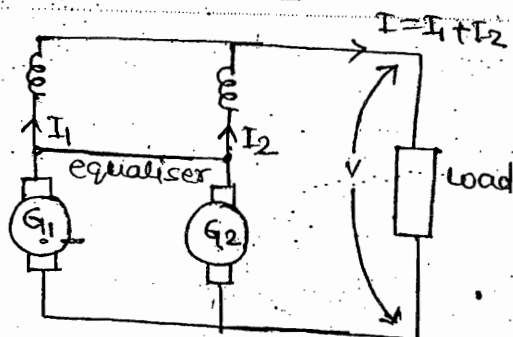
$$I_1 = \frac{E_{g1} - V}{R_{a1}}$$

$$I_2 = \frac{E_{g2} - V}{R_{a2}}$$

*** The load sharing of gen^r operating in parallel is significantly depending on induced emf (directionally proportional)

* In order to have stable or proper parallel operation the voltage c/s should be slightly drooping in nature but not rising.

* Series gen^r in parallel →



$I_2 \uparrow \phi_{se2} \uparrow E_{g2} \uparrow$ (gain load sharing capability)

$I_1 \downarrow E_{g1} \downarrow$ (It goes in motoring mode)

* If we connect an equaliser with the both series gen^r then only it operate in parallel.

* If any one gen^r share more load the increase in its current will increased its flux & induced emf.

* Consequently its load sharing capability will increase in a cumulative manner & that gen^r gets overloaded leaving the other.

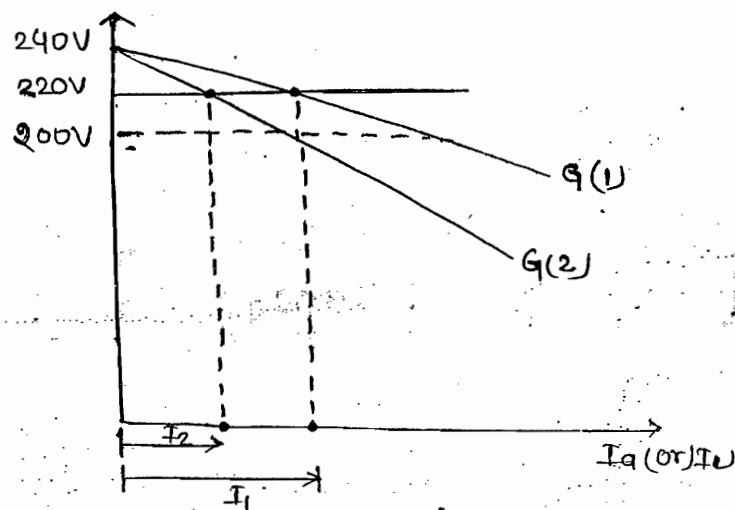
* This is due to the rising c/s of series gen^r.

* In order to make them in parallel an equaliser is required.

* The increased current will bypass into both field wdg to increase there Flux & induced emf equally.

* Equalisers are required for cumulative compound gen^r also.

* Shunt gen^r in parallel →



* Shunt gen^r are best suitable for parallel operation due to there drooping c/s.

* The gen^r which is more drooping will share less load (viceversa)

* By adjusting the c/s the gen^r can be loaded according to there ratings.